

Genetically engineered loaded extracellular vesicles for drug delivery

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Abstract

The use of extracellular vesicles (EVs) for drug delivery is being widely explored by scientists from several research fields. To fully exploit their therapeutic potential, multiple methods for loading EVs have been developed. Although exogenous methods have been extensively utilized, in recent years the endogenous method has gained significant attention. This approach, based on parental cell genetic engineering, is suitable for loading large therapeutic biomolecules such as proteins and nucleic acids. We review the most commonly used EV loading methods and emphasize the inherent advantages of the endogenous method over the others. We also examine the most recent advances and applications of this innovative approach to inform on the diverse therapeutic opportunities that lie ahead in the field of EV-based therapies.